Brijodison

Engineering Notebook Element A

Engineering Service Learning

Problem Statement

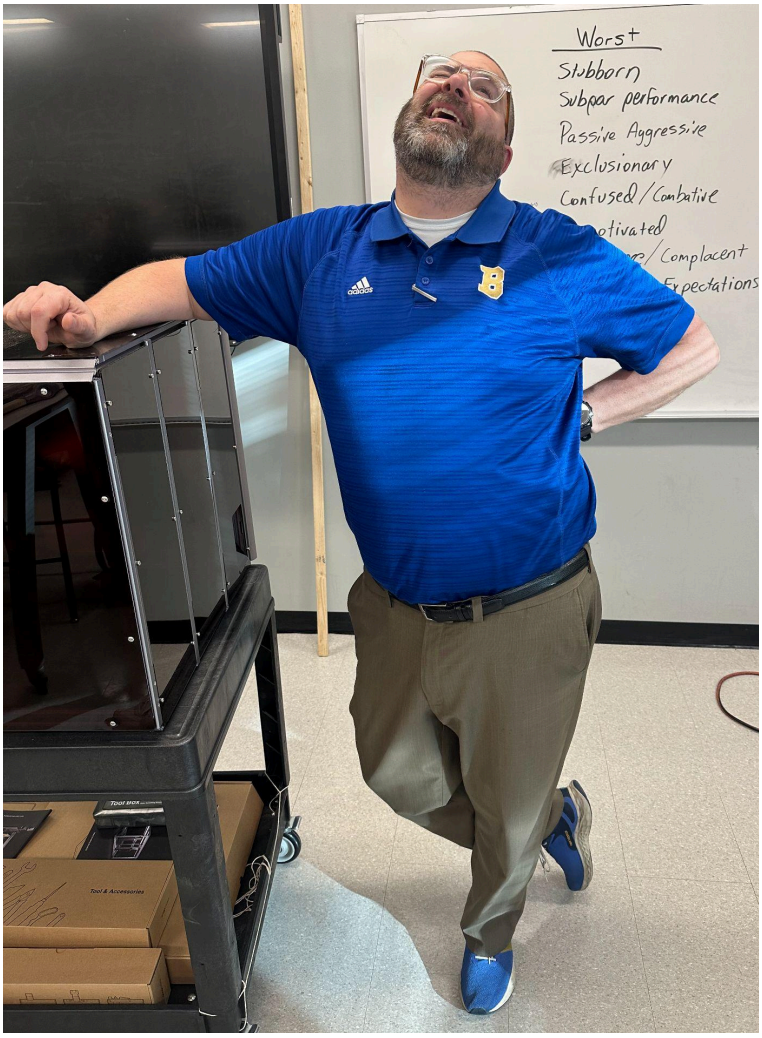
The problem we are trying to solve is whenever Mr. Kiser bends over to push a cart, he experiences pain in his lower back. Mr. Kiser is 6’ 3” with a wingspan of 6’ 3” as well. Not only is he tall enough to where he has to bend down to push the cart, but he also has the L1 and the L2 disc ruptured in his back; this is the main reason for his back pain. The Premier Spine Institute states bending over and back up from a bent position can add pressure to the herniated discs and can bother the surrounding nerves, worsening the symptoms (Premier Spine Institute). While the problem is small-scale, if our solution is formidable then it can solve a large-scale problem. This problem occurs every single time Mr. Kiser has to push this cart, and can become especially more challenging as the size of the load increases.

As I was saying earlier, we want to design our handles to be especially compatible with Mr. Kiser’s multi-purpose cart he uses to push around textbooks, tests and his 3D printer. Once we can be sure it works with Mr. Kiser’s cart, we would like to expand our scopes to compatibility with other carts since many people experience back pain whenever bending over and standing back up. Many people, especially teachers, also use carts in order to move heavy loads from one place to another. According to a study by the National Center for Biotechnology Information (NCBI), musculoskeletal disorders, particularly lower back injuries, are a leading cause of work-related disability among teachers, accounting for nearly 40% of all teacher absences (Pranksy). This shows that even if we are making a cart attachment to solve a problem one person has, it could potentially be the solution for a wide group of people that don’t realize

they need our product. That being said, Mr. Kiser is the only person involved in our problem we are solving for the time being.

Mr. Kiser specifically experiences pain in the upper two discs of his lumbar spine (specifically the L1 and the L2 discs as I stated earlier). This motion places much strain on his already injured lower back. This may lead to even more back pain as well as further injury in the future. The thing that makes Mr. Kiser's problem unique is that this is a pre-existing medical condition that was not caused by pushing a cart whatsoever. He has been using a cart like this throughout his teaching career and hasn't had time to address and act upon this problem until now. Since he is a teacher, he has to follow a specific schedule and does not have time to deal with this since there are more urgent things such as getting to class (with the cart sometimes), as well as teaching many students. Taking all of these factors into consideration, an urgent solution seems more than practical. According to a report from the Harvard Health Blog (HHB), whenever you have limited movement due to chronic pain, it puts stress on your body and mind resulting in pain to worsen (HHB). His pre-existing medical condition can clearly worsen; only reinforcing the fact that the problem must be addressed and furthermore acted upon. According to the Center for Disease Control (CDC), many chronic pains at work have to do with pushing,

pulling, or lifting objects frequently, as well as maintaining these awkward or irregular positions (CDC). To widen the scope even more, Mr. Kiser's problem is especially common for people in the workplace; many not having been derived from pushing a cart in general. Even though Mr. Kiser's injury specifically wasn't caused by work, many people experience the same pain in their workdays and leave it unchecked. In conclusion, having this problem from Mr. Kiser be addressed by a universal solution could be beneficial.



Our primary stakeholders include the students of BHS, Mr. Kiser, any other faculty members, anybody who is at the school to help proctor the ACT, and anybody who has to push a cart. The students of BHS are a primary stakeholder because they may have to push the cart

making their back hurt. They may have to do work for teachers that requires them to push the cart, so we would like to ensure their continued health. This leads to our next primary stakeholder, Mr. Kiser. He will be affected by our device because it will relieve his back pain coming from bending down and back up. Any other teacher or faculty member that pushes the cart will be affected by giving them a better grip and allowing them to maintain comfortable posture while pushing the cart. All ACT proctors will benefit from this because they push the cart one Saturday of every month in order to transport the ACT test booklets. Finally our last primary stakeholder is anybody who pushes a cart at all. It is important that we consider universality while designing our solution in order to widen the scope of the problem and fight against chronic back pain in the workplace as a whole. While a universal cart handle wouldn't exactly do this, it would be a step in the right direction seeing as moving heavy objects would be easier because of the cart, and less painful because of our device. All students of BHS are secondary stakeholders as well because of the time it takes for any tests of textbooks to get to them. They also may have a teacher who is in a bad mood because they hurt their back while bending over to push a supply cart from class to class. Another person who would be a secondary stakeholder would be anybody who has family members who suffer from chronic back pain. The last group of secondary stakeholders include any doctors or medical experts who are fighting against chronic back pain, or even people who think there are workplaces that don't take chronic pain into consideration enough.

This small scoped problem is very important and must be solved now. The CDC states that most of these injuries aren't even caused by an instantaneous movement but wear and tear from poor posture and awkward positions over time (CDC). This is enough reason to justify addressing this problem now rather than later; this is in large part that we can help relieve Mr.

Kiser's back pain now and help prevent any new injuries for our primary stakeholders in the future.

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